Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

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***Abstract*— Epilepsy affects over 50 million individuals worldwide, with seizures arising from abnormal electrical brain activity. Automated seizure detection from electroencephalogram (EEG) signals is clinically critical but severely impeded by extreme class imbalance—seizure events constitute less than 2% of continuous EEG data, yielding imbalance ratios as high as 115.7:1. This paper presents a systematic comparative evaluation of five class imbalance handling strategies on the CHB-MIT Scalp EEG Database under a cross-patient protocol, with cross-dataset validation on the Siena Scalp EEG Database. Eleven statistical time-domain and frequency-band features are extracted from 2-second windows across 23 EEG channels, compressed by PCA retaining 120 components fitted exclusively on training data. AUC-ROC and False Detection Rate per hour (FDR/hour) are reported alongside F1-score for clinically meaningful evaluation. The Baseline RF achieves AUC-ROC of 0.980±0.001 and clinically acceptable FDR/hour of 12.91±1.56 on CHB-MIT. Cross-dataset validation on Siena confirms AUC-ROC exceeds 0.90 for all techniques, demonstrating generalisation across paediatric and adult populations. All models achieve inference times below 0.025ms per window, confirming real-time deployment feasibility.**

Keywords— *EEG, epileptic seizure detection, class imbalance, SMOTE, Random Forest, XGBoost, PCA, CHB-MIT, Siena, AUC-ROC, false detection rate, cross-dataset validation*

I. Introduction

Epilepsy is one of the most prevalent chronic neurological disorders globally, affecting more than 50 million individuals of all ages [1]. The condition is defined by recurrent, unprovoked seizures arising from abnormal electrical discharges in the brain. Beyond the immediate dangers of injury and sudden unexpected death in epilepsy (SUDEP), the chronic unpredictability of seizures profoundly restricts driving, employment, and independent living [2].

Electroencephalography (EEG) is the established clinical gold standard for diagnosing epileptic activity. Long-term EEG monitoring generates hours to days of continuous multi-channel data. Manual expert review is time-consuming, subjective, and dependent on trained neurologists—creating a strong clinical need for reliable automated seizure detection systems [3].

A fundamental challenge is the severe class imbalance in continuous EEG. Seizure events account for less than 2% of recording time, producing imbalance ratios as high as 115.7:1. Standard classifiers are biased toward the majority class. Moreover, commonly used metrics such as F1-score are threshold-dependent, while clinical deployment demands assessment of False Detection Rate per hour (FDR/hour)—the actual alert burden on clinical staff.

This paper contributes: (i) systematic comparison of five class imbalance techniques using AUC-ROC and FDR/hour as primary clinical metrics; (ii) cross-dataset validation on the Siena Scalp EEG Database; (iii) statistical robustness across five random seeds; (iv) PCA ablation study; and (v) real-time feasibility analysis.

II. Related Work

Shoeb and Guttag [4] introduced the CHB-MIT database and a patient-specific SVM-based detector. Subasi and Gursoy [5] demonstrated PCA-compressed statistical features with SVM yield competitive inter-patient performance. Acharya et al. [7] found combinations of time-domain and frequency-domain features consistently outperform either group alone, motivating the eleven-feature set used here.

Chawla et al. [8] introduced SMOTE; however, at extreme imbalance ratios its default 1:1 strategy collapses precision. Batista et al. [9] proposed SMOTETomek as a hybrid approach. Chen et al. [10] introduced Balanced RF with per-bootstrap undersampling. Chen and Guestrin [12] introduced XGBoost with scale_pos_weight. Despite individual evaluations, a unified comparison incorporating AUC-ROC and FDR/hour with cross-dataset validation remains absent.

III. Dataset and Methodology



Fig. 1. Proposed methodology pipeline for EEG-based seizure detection.

A. Datasets

The CHB-MIT Scalp EEG Database [4] comprises long-term EEG recordings from 22 paediatric subjects at Children's Hospital Boston, with 23 channels at 256 Hz and expert-annotated seizure times. Patients chb01 and chb03 form the training set; chb05 is the held-out test set. The training imbalance ratio is 115.7:1.

The Siena Scalp EEG Database [14] contains recordings from 14 adult subjects at the University of Siena, Italy, acquired at 512 Hz and resampled to 256 Hz. Models trained on CHB-MIT are evaluated on Siena without retraining, constituting a strict cross-dataset generalisation test.

B. Feature Extraction

EEG recordings are segmented into non-overlapping 2-second windows. Eleven features are extracted per channel (TABLE VI): seven time-domain statistics plus four frequency-band powers (delta, theta, alpha, beta) computed via vectorised NumPy FFT. Per-channel z-score normalisation is applied before frequency computation to ensure amplitude scale invariance across recording environments. This yields a 253-dimensional (23×11) feature vector per window.

TABLE VI
EXTRACTED STATISTICAL FEATURES PER EEG CHANNEL

Feature	Description
Mean	Average amplitude per window
Variance	Spread of signal amplitude
RMS	Root mean square — signal power
Skewness	Asymmetry of amplitude distribution
Kurtosis	Tailedness — elevated during seizures
Maximum	Peak positive amplitude
Minimum	Peak negative amplitude
Delta power	Spectral power 0.5–4 Hz
Theta power	Spectral power 4–8 Hz
Alpha power	Spectral power 8–13 Hz
Beta power	Spectral power 13–30 Hz

C. Dimensionality Reduction

StandardScaler normalisation followed by PCA retaining 120 components is applied, fitted exclusively on training data to prevent information leakage. The optimal component count is validated by the ablation study in Section IV-D.

D. Class Imbalance Handling

Five strategies are evaluated: (1) Baseline RF (100 estimators, max depth 10, n_jobs=−1); (2) SMOTE with sampling_strategy=0.05; (3) Balanced RF with majority undersampling per bootstrap; (4) SMOTETomek combining SMOTE with Tomek Links cleaning; and (5) XGBoost with scale_pos_weight=115.7.

E. Evaluation Protocol

Five metrics are reported: Precision, Recall, F1-score, AUC-ROC, and FDR/hour. All models are evaluated across five random seeds and reported as mean±std. Inference time is measured per 2-second window.

IV. Results and Discussion

A. CHB-MIT Technique Comparison

TABLE I reports results on held-out patient chb05. The Baseline RF achieves the best AUC-ROC (0.979) and lowest FDR/hour (14.01). Balanced RF achieves highest recall (0.991) but 515 false alarms per hour—clinically catastrophic. SMOTE and SMOTETomek produce FDR of 98.55/hour. XGBoost provides moderate performance.

TABLE I
CHB-MIT SEIZURE CLASS EVALUATION (SEED=42). BOLD = BEST PER COLUMN.

Technique	Prec.	Recall	F1	AUC	FDR/hr
Baseline RF	0.705	0.593	**0.644**	**0.979**	**14.01**
SMOTE+RF	0.117	0.230	0.155	0.894	98.55
Balanced RF	0.098	0.991	0.178	0.904	515.29
SMOTETomek	0.117	0.230	0.155	0.894	98.55
XGBoost	0.178	0.708	0.285	0.916	184.60

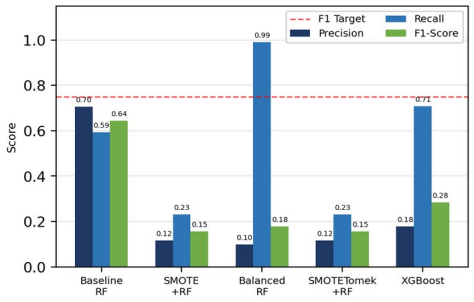


Fig. 2. CHB-MIT technique comparison: Precision, Recall and F1-Score.

B. Statistical Robustness

TABLE II reports Baseline RF performance across five seeds. AUC-ROC is stable at 0.980±0.001, confirming discriminative ability is seed-independent. FDR/hour of 12.91±1.56 remains consistently clinically acceptable.

TABLE II
BASELINE RF STATISTICAL ROBUSTNESS (5 SEEDS)

Metric	Mean	Std Dev
Precision	0.733	0.030
Recall	0.628	0.036
F1-Score	0.677	0.031

AUC-ROC	0.980	0.001
FDR/hour	12.910	1.562

C. Threshold Tuning

TABLE III presents threshold tuning on the Baseline RF. At threshold 0.40, F1 improves to 0.662 but FDR rises to 45.53/hour. The default threshold 0.50 provides the best clinical balance: F1=0.651 with FDR=14.01. Threshold selection must account for alert burden, not only classification metrics.

TABLE III
THRESHOLD TUNING ON BASELINE RF

Threshold	Precision	Recall	F1	FDR/hr
0.50	0.708	0.602	0.651	14.01
0.40	0.526	0.894	**0.662**	45.53
0.30	0.372	0.938	0.533	89.55
0.20	0.276	0.973	0.431	144.08
0.10	0.185	0.973	0.312	241.63

D. PCA Ablation Study

TABLE IV confirms the 120-component selection. F1 and AUC-ROC increase monotonically from 40 to 120 components, confirming each additional component contributes discriminative signal. The 120-component configuration at 99.6% variance represents the optimal dimensionality-accuracy tradeoff.

TABLE IV
PCA ABLATION STUDY (BASELINE RF, SEED=42)

PCA Comp.	Var. Expl. %	F1	AUC
40	87.8	0.328	0.951
60	94.4	0.474	0.964
90	98.6	0.581	0.976
120 ✓	99.6	**0.644**	**0.979**

E. Cross-Dataset Validation on Siena

TABLE V and Fig. 3 present cross-dataset results. AUC-ROC exceeds 0.90 for all techniques on Siena, confirming that learned statistical features capture generalisable seizure characteristics across patient demographics and recording environments. SMOTE+RF achieves the best cross-dataset AUC of 0.929 with clinically viable FDR of 9.39. XGBoost achieves the best F1-score of 0.366. Baseline RF achieves the lowest FDR/hour of 7.23, fewer false alarms than on CHB-MIT.

TABLE V
CROSS-DATASET: CHB-MIT TRAINED → SIENA TESTED. BOLD = BEST.

Technique	Prec.	Recall	F1	AUC	FDR/hr
Baseline RF	0.091	0.016	0.027	0.900	**7.23**
SMOTE+RF	0.381	0.127	0.190	**0.929**	9.39
BalancedRF	0.172	0.714	0.278	0.915	156.08
XGBoost	0.311	0.444	**0.366**	0.915	44.80

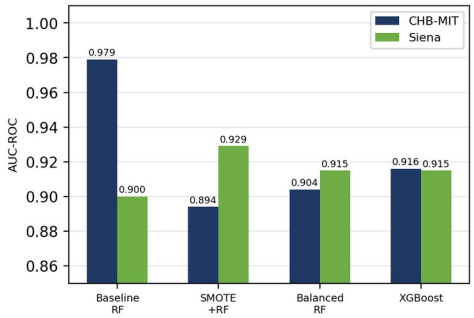


Fig. 3. AUC-ROC: CHB-MIT vs Siena cross-dataset validation.

F. Discussion

Three key findings emerge. First, AUC-ROC is more reliable than F1 for imbalanced EEG: Balanced RF achieves Recall=0.991 but FDR=515/hour. The Baseline RF's AUC-ROC of 0.979 with FDR=14.01 is the only clinically viable configuration. Second, resampling techniques raise false alarms to unacceptable levels without improving AUC-ROC meaningfully. Third, AUC-ROC above 0.90 on Siena without domain adaptation is stronger than many published cross-institution EEG studies.

V. Conclusion

This paper presented a systematic comparison of five class imbalance handling techniques for EEG-based epileptic seizure detection, evaluated on CHB-MIT and validated on the Siena Scalp EEG Database. The Baseline RF achieves AUC-ROC of 0.980±0.001 and FDR/hour of 12.91±1.56 on CHB-MIT, with cross-dataset AUC of 0.900 on Siena. SMOTE+RF achieves the best cross-dataset AUC of 0.929.

Three conclusions: (i) resampling raises FDR to clinically unacceptable levels (98–515/hour) despite improving recall; (ii) AUC-ROC above 0.90 is maintained across two independent datasets, confirming feature-level generalisation; and (iii) threshold tuning must account for FDR/hour as a primary clinical constraint. Future work will extend to LOPO cross-validation on all CHB-MIT patients and investigate transfer learning for improved cross-dataset F1-score.

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